
AR 510: Underactuated Robotics

Course Code	AR 510
Course Name	Underactuated Robotics
L-T-P-C	3-0-0-3
Prerequisites	Consent of faculty advisor
Intended for	UG, PG and PhD
Mutual Exclusion	None
Approval	50th BoA (Jul 2023)

Course Contents

- **Introduction to Underactuated Robotics:** Motivation, Fully-actuated vs Underactuated systems, Feedback equivalence, Input and State constraints. (6 hours)
- **Robot dynamics and model-based control:** Nonlinear dynamics with a constant torque, Acrobats, Cart-poles, and Quadrotors. (6 hours)
- **Dynamic programming:** Lyapunov analysis; Trajectory optimization, Trajectory stabilization; Policy search; Multibody parameter estimation, Formulating control design as an optimization. (6 hours)
- **Simple models of walking and running:** Limit cycles, Simple models of walking such as Rimless wheel, Compass gait, Knead walker, Curved feet, Simple models of running such as Spring-loaded inverted pendulum (SLIP), Hopping robots, Toward human-like running. (8 hours)
- **Planning and control through contact:** Sampling-based motion planning, Complete motion planning, Feedback motion planning, Planning as Combinatorial + Continuous Optimization. (8 hours)
- **Stochastic dynamics:** Stochastic/ robust control, Master equation, Stationary distribution, Costs and constraints for stochastic systems, Finite Markov Decision Processes, From linear models to deep models. (8 hours)

Text Books

1. Tedrake R., **Underactuated Robotics: Algorithms for Walking, Running, Swimming, Flying, and Manipulation**, MIT Press.

References

1. Xin X. and Liu Y., **Control Design and Analysis for Underactuated Robotic Systems**, Springer.
2. Birglen L., Laliberte T., and Gosselin C., **Underactuated Robotic Hands**, Springer.